Polarization (Griffiths §4.2): structured notes

Griffiths, *Introduction to Electrodynamics* (5th ed.), pp. 173–180. Full content of the study notes, arranged by topic for revision. Equation numbers match the book. Figures are sketches, not the book's figures.

Before you start

- Potential of a single dipole, $V = \dfrac{1}{4\pi\epsilon_0}\dfrac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}$ (Eq. 4.8, from §4.1).
- Polarization $\mathbf{P}$ = dipole moment per unit volume (§4.1.4). Field of a uniformly polarized sphere (Eq. 4.14 is derived below).
- Surface and volume charge densities, and the potential they produce (book Eqs. 3.86–3.87 for a surface charge $\cos\theta$ distribution on a sphere).
- Maths: gradient ∇, divergence $\nabla\cdot$, product rule for ∇, integration by parts, the divergence theorem. Partial derivatives are needed (not in Class 12 maths).

1. Potential of a polarized object (§4.2.1)

1.1 Starting point: sum over dipoles

- Each volume element $d\tau'$ at $\mathbf{r}'$ carries a dipole $\mathbf{P}(\mathbf{r}')\,d\tau'$.
- Summing the dipole potentials of Eq. 4.8 over the volume:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int_V \frac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\mathcal{R}}}}{\mathcal{R}^2}\,d\tau' \quad (4.9).$$

- $\boldsymbol{\mathcal{R}} = \mathbf{r} - \mathbf{r}'$ is the vector from the source point to the field point. This is correct in principle but not useful for calculation.

1.2 Rewrite using a gradient

- Key identity (differentiation with respect to the source coordinates $\mathbf{r}'$):

$$\nabla'\left(\frac{1}{\mathcal{R}}\right) = \frac{\hat{\boldsymbol{\mathcal{R}}}}{\mathcal{R}^2}.$$

- Substituting:

$$V = \frac{1}{4\pi\epsilon_0}\int_V \mathbf{P}\cdot\nabla'\left(\frac{1}{\mathcal{R}}\right)d\tau'.$$

1.3 Integrate by parts

- Product rule: $\nabla'\cdot\left(\dfrac{\mathbf{P}}{\mathcal{R}}\right) = \dfrac{1}{\mathcal{R}}\nabla'\cdot\mathbf{P} + \mathbf{P}\cdot\nabla'\left(\dfrac{1}{\mathcal{R}}\right)$. Rearranged:

$$V = \frac{1}{4\pi\epsilon_0}\left[\int_V \nabla'\cdot\left(\frac{\mathbf{P}}{\mathcal{R}}\right)d\tau' - \int_V \frac{1}{\mathcal{R}}(\nabla'\cdot\mathbf{P})\,d\tau'\right].$$

- Divergence theorem on the first integral turns it into a surface integral:

$$V = \frac{1}{4\pi\epsilon_0}\oint_S \frac{1}{\mathcal{R}}\,\mathbf{P}\cdot d\mathbf{a}' - \frac{1}{4\pi\epsilon_0}\int_V \frac{1}{\mathcal{R}}(\nabla'\cdot\mathbf{P})\,d\tau' \quad (4.10).$$

1.4 Define the bound charges

DEFINITIONS
Bound surface charge: $\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}}$ (4.11), where $\hat{\mathbf{n}}$ is the outward normal.
Bound volume charge: $\rho_b = -\nabla\cdot\mathbf{P}$ (4.12).

- Substituting into Eq. 4.10 gives

RESULT
$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\oint_S \frac{\sigma_b}{\mathcal{R}}\,da' + \frac{1}{4\pi\epsilon_0}\int_V \frac{\rho_b}{\mathcal{R}}\,d\tau'$ (4.13).

- Meaning: the potential, and so the field, of a polarized object equals that of a volume charge ρ_b plus a surface charge σ_b.
- Method: find σ_b and ρ_b first, then use Gauss's law or the usual potential integrals for those charges.

2. Worked example: uniformly polarized sphere (Example 4.2)

- Radius R, polarization $\mathbf{P}$ uniform. Choose the z-axis along $\mathbf{P}$.
- Volume charge: $\rho_b = -\nabla\cdot\mathbf{P} = 0$, since $\mathbf{P}$ is uniform.
- Surface charge: $\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}} = P\cos\theta$.
- So the problem is the field of a $P\cos\theta$ surface charge on a sphere. The book uses the known potential of that distribution (Eqs. 3.86–3.87):

$$V(r,\theta) = \begin{cases} \dfrac{P}{3\epsilon_0}\,r\cos\theta & r \le R, \\[2ex] \dfrac{P}{3\epsilon_0}\dfrac{R^3}{r^2}\cos\theta & r \ge R. \end{cases}$$

- Inside: $r\cos\theta = z$, so the potential is linear in z and the field is uniform:

RESULT INSIDE
$\mathbf{E} = -\nabla V = -\frac{P}{3\epsilon_0}\hat{\mathbf{z}} = -\frac{1}{3\epsilon_0}\mathbf{P}$ for $r < R$ (4.14).

- Outside: the potential equals that of a perfect dipole at the origin, with moment equal to the total dipole moment of the sphere:

RESULT OUTSIDE
$V = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}$ for $r \ge R$ (4.15), with $\mathbf{p} = \frac{4}{3}\pi R^3\,\mathbf{P}$ (4.16).

- Check: $\dfrac{p}{4\pi\epsilon_0} = \dfrac{(4/3)\pi R^3 P}{4\pi\epsilon_0} = \dfrac{R^3 P}{3\epsilon_0}$, which matches the outside potential above.

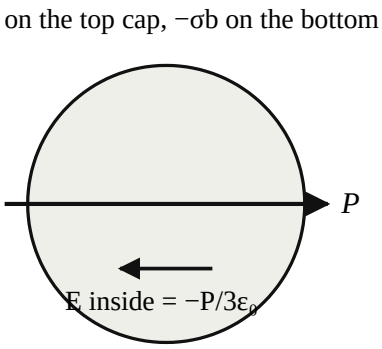


Figure A. My sketch of the Example 4.2 sphere: uniform P inside, bound surface charge on the surface.

3. Physical meaning of bound charges (§4.2.2)

3.1 Dipole chain

- Take a line of dipoles. Inside the line, the head of one dipole cancels the tail of its neighbour.
- At the two ends, one charge is left over: positive at the right end, negative at the left.
- Such leftover charge is called **bound charge**. It cannot be removed, because each electron stays attached to its own atom or molecule. Otherwise it behaves like any other charge.

3.2 Amount of bound charge at a flat end

- Consider a tube of dielectric parallel to $\mathbf{P}$, cross-section A. Each chunk of length d has dipole moment $P(Ad)$. Written as qd, the charge at the end of the tube is

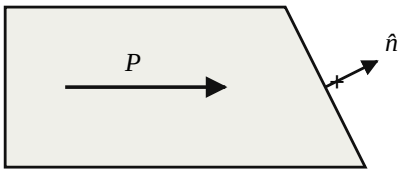
$$q = PA.$$

- For a flat cut perpendicular to $\mathbf{P}$: $\sigma_b = q/A = P$.
- For an oblique cut at angle θ to the perpendicular, the end area is $A_{\text{end}} = A/\cos\theta$. The charge is unchanged, so

RESULT

$$\sigma_b = \frac{q}{A_{\text{end}}} = P\cos\theta = \mathbf{P}\cdot\hat{\mathbf{n}}.$$

- This agrees with the rigorous result of §4.2.1 (Eq. 4.11), and now we know where the charge comes from.



σb = P cos θ = P·n̂ on the slanted end

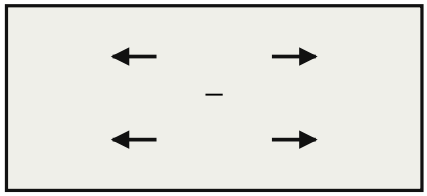
Figure B. My sketch of an oblique cut of a polarized tube (§4.2.2).

3.3 Bound volume charge from non-uniform P

- If $\mathbf{P}$ varies in space, charge also piles up inside the material.
- Where $\mathbf{P}$ diverges (spreads out), there is a net negative pile-up, because the flux of $\mathbf{P}$ leaving a volume leaves a deficit of positive charge behind.
- Total bound charge in a volume must equal minus the flux of $\mathbf{P}$ out of it:

$$\int_V \rho_b \, d\tau = -\oint_S \mathbf{P} \cdot d\mathbf{a} = -\int_V (\nabla \cdot \mathbf{P}) \, d\tau.$$

- Since this holds for any volume: $\rho_b = -\nabla \cdot \mathbf{P}$ (4.12), confirming §4.2.1.



P diverges, so ρb = −∇·P < 0 in the middle

Figure C. My sketch of a diverging P: net negative bound charge builds up inside.

3.4 Example 4.3: two overlapping spheres

- A uniformly polarized sphere can be seen as a positive sphere and a negative sphere of equal size, displaced by $\mathbf{d}$. Before polarization they cancel exactly.
- Polarization moves the positive charges up and the negative charges down. Two caps of uncancelled charge remain: positive at the top, negative at the bottom. These are the bound surface charges σ_b.
- Inside the overlap, the field of the two uniform spheres is (from Prob. 2.19):

$$\mathbf{E} = -\frac{1}{4\pi\epsilon_0}\frac{q\,\mathbf{d}}{R^3}.$$

- With $\mathbf{p} = q\mathbf{d} = \frac{4}{3}\pi R^3 \mathbf{P}$, this is $\mathbf{E} = -\frac{1}{3\epsilon_0}\mathbf{P}$, which matches Eq. 4.14.

- Outside: each sphere acts as a point charge at its centre, so the total is a dipole with potential $\frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}$. This matches Eq. 4.15.
- Note: $\mathbf{d}$ is a small fraction of an atomic radius. The book's figure exaggerates it.

4. Macroscopic and microscopic fields (§4.2.3)

Background. Book footnote 5 says this subsection can be skipped without loss of continuity. It is not in the syllabus list for Module I.

4.1 Why the method is questioned

- Section 1 used perfect (point) dipoles and replaced a set of discrete molecular dipoles by a continuous density $\mathbf{P}$.
- Outside the dielectric, the molecules are far away, so the dipole approximation holds and the method is safe.
- Inside, the actual field is very complicated on the atomic scale. It is huge near an electron and small or reversed a short distance away, and it changes with time.

4.2 Macroscopic field

- Definition: the average of the true microscopic field over a region that is large enough to contain very many atoms, and small enough not to smooth out large-scale variations. In practice, the averaging region is much smaller than the object.
- Ordinarily, "the field inside matter" means this macroscopic field. It is also the basis of the idea of density of a material.

4.3 Split the field at a point

- Draw a small sphere about the point $\mathbf{r}$. The macroscopic field splits into two parts:

$$\mathbf{E} = \mathbf{E}_{\text{out}} + \mathbf{E}_{\text{in}}.$$

- $\mathbf{E}_{\text{out}}$: the average field over the sphere due to charges outside it. The average over a sphere due to outside charges equals the field at the centre (Prob. 3.51d). Far enough away, Eq. 4.9 applies:

$$V_{\text{out}} = \frac{1}{4\pi\epsilon_0}\int_{\text{outside}} \frac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\mathcal{R}}}}{\mathcal{R}^2}\,d\tau' \quad (4.17).$$

- $\mathbf{E}_{\text{in}}$: the average field due to charges inside the sphere. By Eq. 3.105 it depends only on the total dipole moment:

$$\mathbf{E}_{\text{in}} = -\frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}}{R^3} = -\frac{1}{3\epsilon_0}\mathbf{P} \quad (4.18).$$

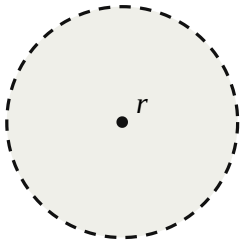
4.4 Why this gives the same answer as §4.2.1

- $\mathbf{P}$ does not vary much over the small sphere. So the part of the integral over the small sphere equals the field at the centre of a uniformly polarized sphere, which is $-\frac{1}{3\epsilon_0}\mathbf{P}$ (Eq. 4.14).
- Eq. 4.18 supplies exactly this term, so it is put back in.
- Result: the macroscopic potential is the integral over the whole dielectric:

RESULT

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\mathcal{R}}}}{\mathcal{R}^2}\,d\tau' \quad (4.19).$$

- This is the same integral as Eq. 4.9. The method of §4.2.1 therefore gives the macroscopic field inside the dielectric.
- The conclusion does not depend on the shape of the averaging region, though the book uses a sphere. A cube or ellipsoid would give the same final answer, with more difficult algebra.



E = E_out + E_in: outside charges and inside dipoles

Figure D. My sketch of the averaging sphere (§4.2.3). Charges outside the sphere give E_out; the dipoles inside give E_in.

5. What you must be able to do

- Derive Eq. 4.10 from Eq. 4.9, using the gradient identity, the product rule and the divergence theorem.
- State σ_b and ρ_b and the potential Eq. 4.13.
- Find the field inside and outside a uniformly polarized sphere (Eqs. 4.14–4.16).
- Explain why an oblique cut carries $\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}}$ (§3.2).
- Explain why a diverging $\mathbf{P}$ produces negative bound charge (§3.3).
- Check that the total bound charge of a neutral dielectric is zero (Prob. 4.14).

6. Problems in the excerpt (not solved here)

- Prob. 4.10: $\mathbf{P} = k\mathbf{r}$ in a sphere; find σ_b, ρ_b, and the fields.
- Prob. 4.11: bar electret (frozen-in uniform P in a short cylinder); sketch the field for three cases.

- Prob. 4.12: compute the potential of the uniformly polarized sphere directly from Eq. 4.9.
- Prob. 4.13: field inside and outside a long cylinder with P perpendicular to its axis.
- Prob. 4.14: prove from Eqs. 4.11 and 4.12 that the total bound charge vanishes.

7. Open points

- The sign and form of the outside field of the sphere (Eq. 4.15) were checked by the potential only. The field outside has not been computed separately here.
- Eq. 3.105 (average field of the dipoles inside a sphere) and Prob. 3.51d are cited from the book. Their proofs are not reproduced in these notes.
- Problems 4.10–4.13 are listed, not solved.